A Simulation of Constraint-Induced Structural Regimes in a Perishable Supply Network

1. Introduction

This paper presents a constructed simulation of a perishable agricultural supply network (potatoes) as an application of a field-based constraint dynamics framework. The objective is not to extend the underlying theory, but to demonstrate how its operators manifest in a controlled system where demand is held constant and all observed behaviors arise from structural constraint interaction.

The simulation isolates the emergence of three distinct regimes:

- Accumulation under localized constraint
- Functional isolation under interacting constraints
- Recovery via elasticity under expanded buffering and routing capacity

The results show that system behavior is governed by constraint structure rather than parameter tuning or exogenous demand variation. In particular, the simulation demonstrates that topological connectivity is insufficient to ensure functional transmissibility, and that system failure may arise in fully connected networks through downstream acceptance limitations alone.

2. Mapping Simulation to Field-Based Framework

The simulation instantiates the operators defined in the field-based specification (Castle, 2026a) and the networked constraint formulation (Castle, 2026b).

Each node (i) carries observable state—inventory (age-bucketed), backlog, throughput, and spoilage—which map to:

- Stress ($S_{i,t}$): realized burden from saturation, backlog persistence, and age concentration
- Elasticity ($E_{i,t}$): buffering capacity via storage, routing flexibility, and clearance
- Leaked stress ($L_{i,t}$): unresolved constraint exceeding absorption
- Reaction ($R_{i,t}$): system response (rerouting, rejection, spoilage)

Edges encode constrained interaction with capacity, delay, headroom ($H_{ij,t}$), openness ($O_{ij,t}$), and destination acceptance ($A_{j,t}$), forming an effective transmissibility operator consistent with (Castle, 2026a; Castle, 2026b).

Propagation operates on leaked stress gradients, following the leaked-stress transport formulation, ensuring that only transmissible constraint—not total stress—is transported. Persistence ($q_{i,t}$) governs accumulation, as required for retained accumulation under constrained interaction (Castle, 2026b).

3. Simulation Construction

The simulation is calibrated using published U.S. agricultural and freight datasets, with all model constants derived from documented sources. Production levels are scaled from USDA National Agricultural Statistics Service (NASS) state-level potato output (USDA NASS, 2025), while demand is constructed from USDA Economic Research Service (ERS) per-capita consumption data (USDA ERS, 2022). Transport capacities and transit characteristics are derived from the Freight Analysis Framework (FHWA, 2022; BTS & FHWA, 2025), and storage behavior and spoilage dynamics are parameterized using USDA Agricultural Research Service storage guidelines (USDA ARS, 2016) and PACA good delivery standards (USDA AMS, n.d.).

To isolate structural effects, all experiments are run under a fixed demand and production profile with no stochastic variation or exogenous shocks. Model scale is uniformly reduced to a sub-regional corridor (approximately 1.9% of observed flows) to maintain steady-state operation under baseline conditions. As a result, differences across experiments arise from constraint interaction and network structure, rather than from variation in input data or parameter tuning.

Derived quantities (S, L, E, P) are computed from these observables as observable projections of underlying constraint structure (Castle, 2026c). Demand is fixed across all experiments.

4. Experimental Design

Four experiments isolate structural effects:

- **Exp1 (Baseline):** No constraints
- **Exp2 (Single):** Storage constraint
- **Exp3 (Multi):** Storage + processing + route constraints
- **Exp4 (Elasticity):** Added storage and alternate routing

All experiments operate under identical demand and production conditions derived from the calibrated baseline.

5. Results

5.1 Leaked Stress Dynamics

Single constraints produce localized increases in leaked stress, while interacting constraints suppress transmission and redistribute unresolved burden. Elasticity restores flow without eliminating stress magnitude.

Leaked stress reflects transmissibility failure, not just burden intensity (see Fig. 1).

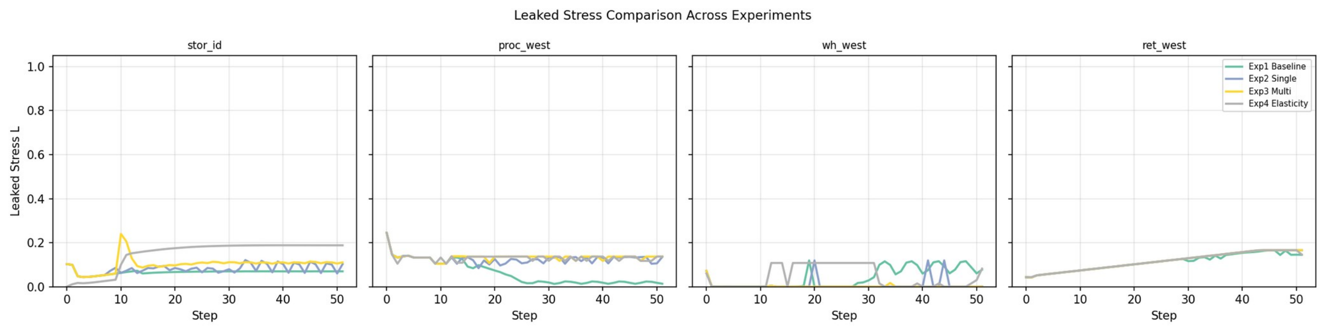

Figure 1: Leaked stress L across experiments. Differences arise despite identical demand, indicating constraint-driven transmissibility variation.

5.2 Accumulation Patterns (Near-Expiry Concentration)

Under storage constraint, age concentration stabilizes locally. Under interacting constraints, accumulation spreads upstream. Elasticity diffuses age pressure across the network.

Accumulation requires persistence and constrained outflow (see Fig.2).

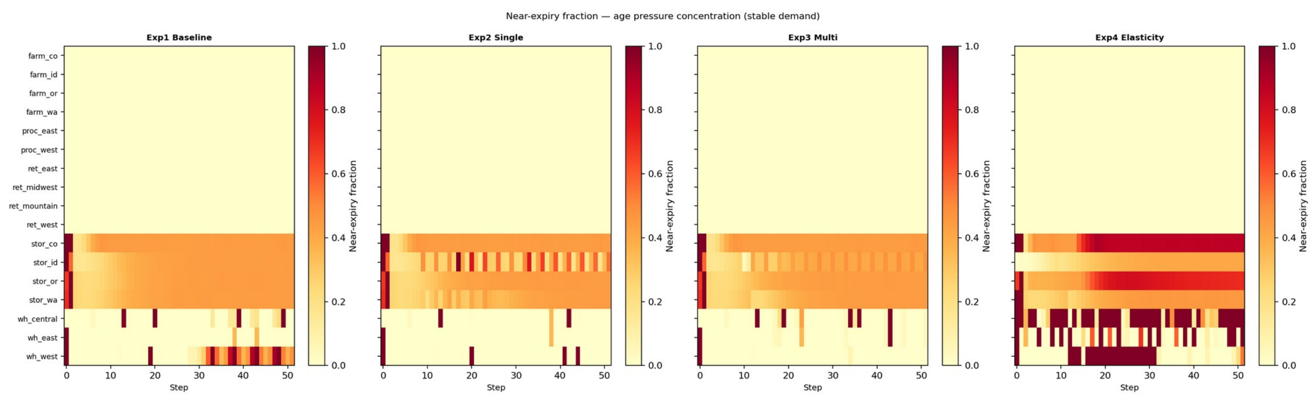

Figure 2: Near-expiry concentration by node. Accumulation emerges under constrained outflow and persists over time.

5.3 Connectivity vs Transmissibility

Physical connectivity remains effectively complete across all experiments. However, usable transmission capacity declines under constraint (see Fig. 3).

The system remains connected while becoming non-resolving, consistent with function isolation arising under constrained transmissibility rather than topological disconnection (Castle, 2026a; Castle, 2026b).

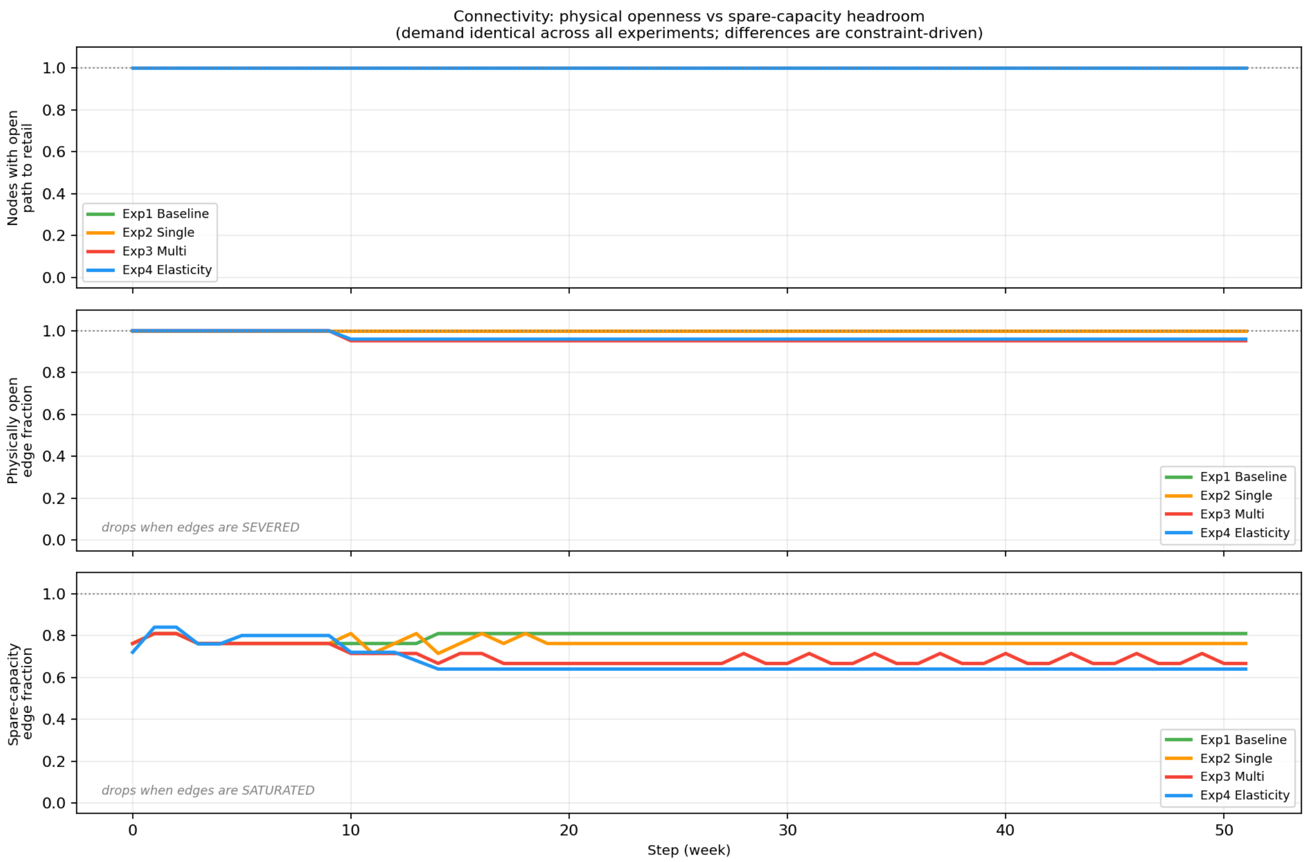

Figure 3: Physical openness vs effective transmissibility. The network remains fully connected while functional capacity degrades under constraint.

5.4 Regime Classification

Distinct regimes emerge across experiments despite identical demand (see Fig. 4):

- Baseline: dispersal
- Single constraint: accumulation
- Multi-constraint: isolation
- Elasticity: recovery

Isolation occurs without graph disconnection.

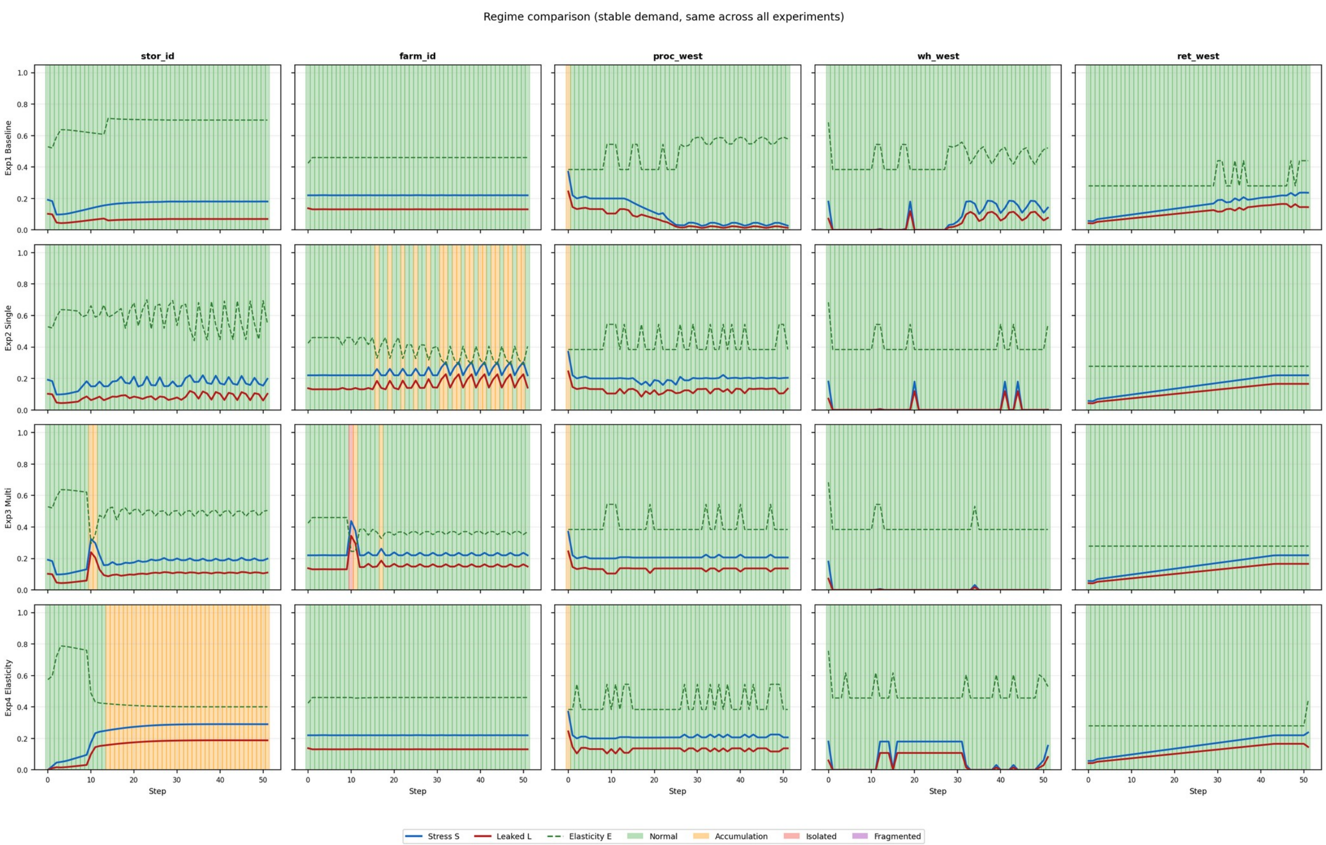

Figure 4: Regime classification across nodes and experiments. Distinct structural regimes arise from constraint interaction despite identical demand.

5.5 Throughput Decomposition

Exp2 — Single Constraint

Blocked flow dominates under storage limitation, but downstream acceptance remains largely intact and clearance continues (see Fig. 5), as implied by receiver-limited transmission in the field formulation (Castle, 2026a).

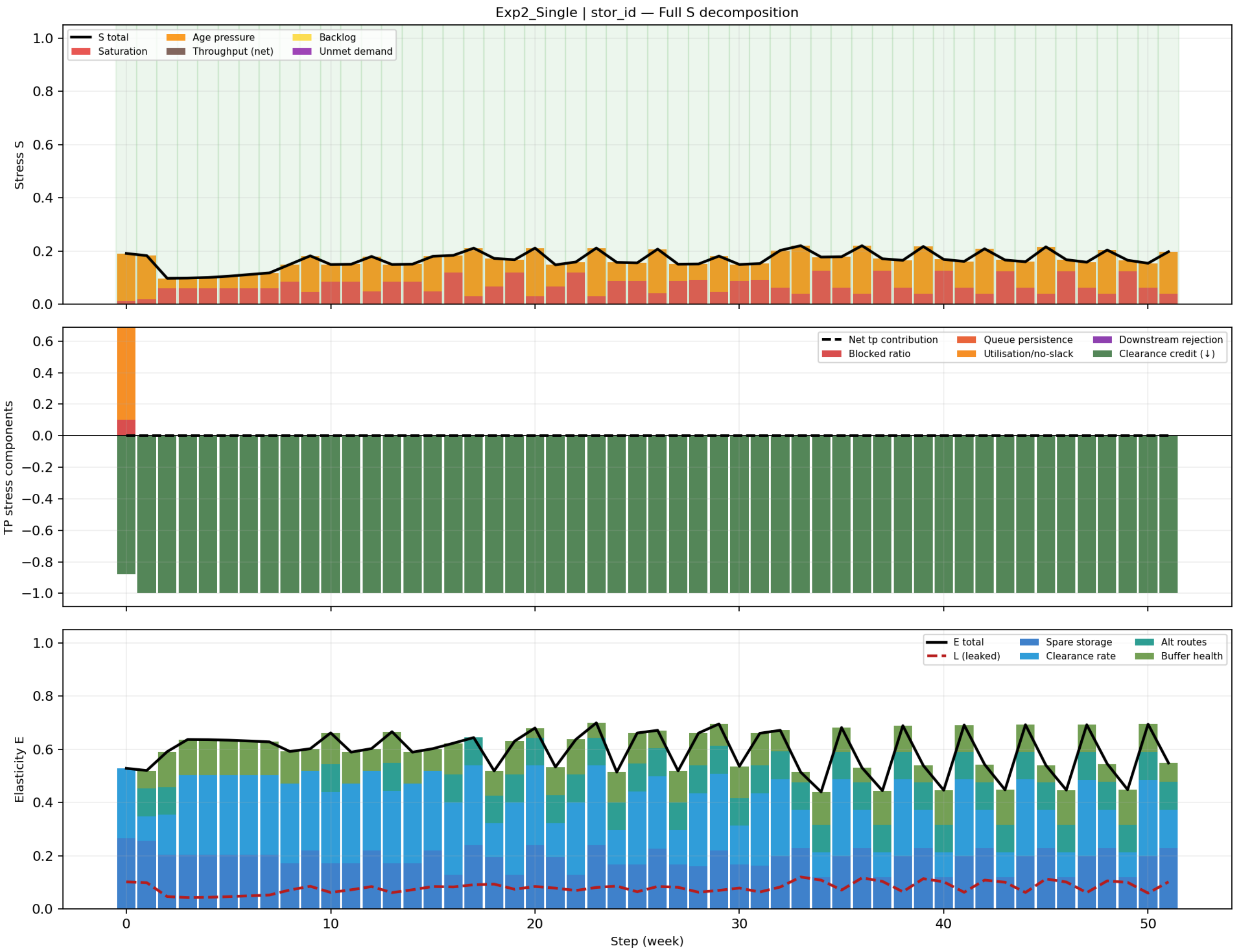

Figure 5: Throughput decomposition under single constraint. Accumulation occurs through blocked flow while clearance remains functional.

Exp3 — Multi-Constraint

Acceptance failure increases sharply under interacting constraints, reducing effective throughput despite open paths (see Fig. 6).

Downstream acceptance is the primary mechanism of functional isolation.

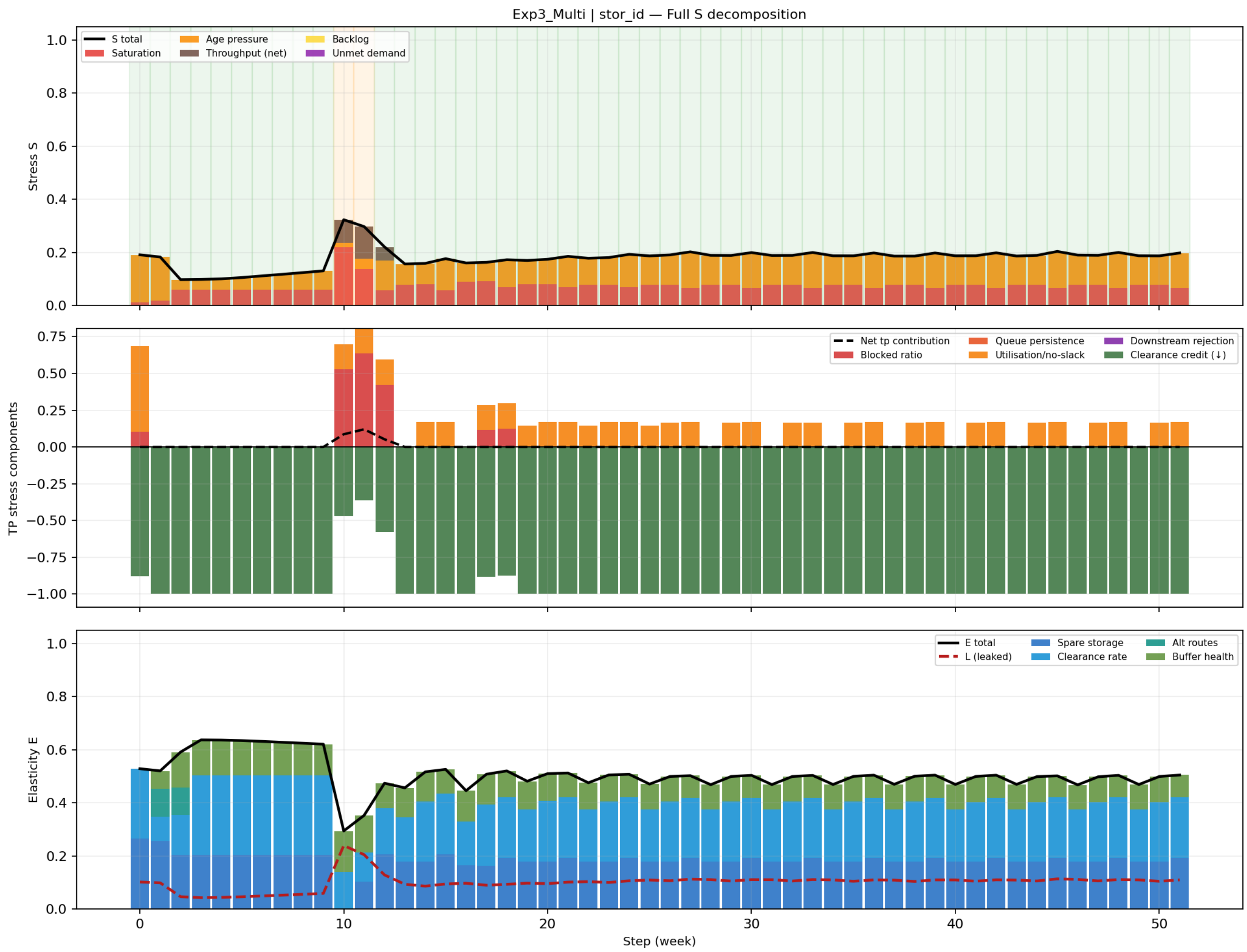

Figure 6: Throughput decomposition under multi-constraint conditions. Acceptance failure, not edge removal, drives collapse in transmissibility.

6. Mechanism Analysis

6.1 Single Constraint to Accumulation

When outflow is restricted by storage capacity, inflow exceeds clearance and backlog persists. This produces sustained leaked stress in queue and age components, as shown in the near-expiry concentration patterns. Previous observations show this as accumulation under constrained outflow and persistence (Castle, 2026b).

6.2 Multi-Constraint to Functional Isolation

Interacting constraints reduce headroom and acceptance simultaneously. Even with physically open edges, effective transmissibility collapses, corresponding to dynamic isolation under state-dependent interaction (Castle, 2026b). This is reflected in the throughput decomposition where acceptance failure dominates.

6.3 Acceptance Failure as Isolation Mechanism

Isolation arises from downstream rejection rather than edge removal, known as acceptance-modulated propagation (Castle, 2026a).

6.4 Elasticity to Recovery Without Stress Reduction

Elasticity increases buffering and routing flexibility, restoring clearance and transmissibility. This is in line with elasticity as a dynamic damping and recovery mechanism (Castle, 2026c). Leaked stress declines, but total stress remains elevated, indicating recovery through improved resolution rather than reduced burden.

7. Structural Interpretation

System behavior is determined by constraint interaction across three mechanisms:

- Local damping via elasticity
- Constraint-induced retention via persistence
- Network transport of leaked stress

(Castle, 2026a)

Transmissibility depends jointly on openness, headroom, and acceptance as specified in the effective interaction operator (Castle, 2026a; Castle, 2026b). Functional isolation occurs when these conditions fail, even in fully connected networks. Clearance operates as the primary damping channel, restoring system function without eliminating underlying stress, through its contribution to elasticity rebuilding (Castle, 2026c).

8. Conclusion

The simulation demonstrates that accumulation, isolation, and recovery emerge as structural properties of constrained interaction, not as artifacts of demand variation or parameter selection.

Single constraints produce localized accumulation through persistence and restricted outflow. Interacting constraints produce functional isolation through acceptance failure without requiring graph disconnection. Elasticity restores clearance and transmissibility without necessarily reducing stress magnitude.

These results confirm that system dynamics are governed by constraint structure and operator interaction, consistent with the field-based interpretation of constraint dynamics, providing a concrete demonstration of field-based constraint dynamics in a perishable supply network.

References

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